

Fourier series

are infinite series designed to represent general periodic function in terms of simple one, namely, cosine and sine.

Periodic function

A function $f(x)$ is said to be periodic if there exist a number $p > 0$, s.t

$$f(x+p) = f(x)$$

$$P = 2L$$

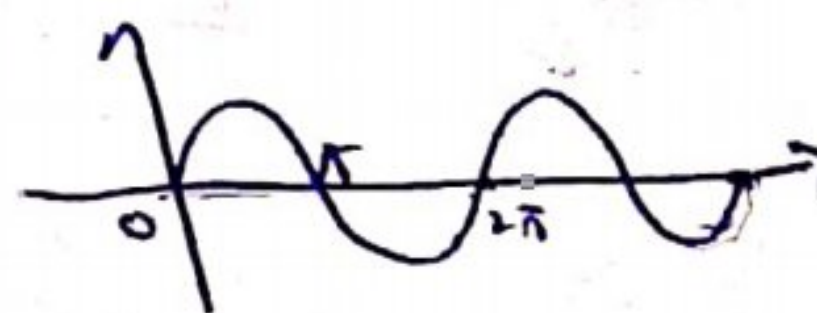
$$\frac{P}{2} = L$$

where p is called period of $f(x)$.

Ex

$$\sin(x+2\pi) = \sin x$$

$$\cos(x+2\pi) = \cos x$$



Fourier series

Fourier series

If f is a piecewise continuous function on $[-L, L]$, its Fourier series is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right) \quad (1)$$

where Fourier coefficients are,

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx \quad (2)$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \quad (3)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \quad (4)$$

where $f(x+P) = f(x) \quad (5)$

$$\Rightarrow P = 2L \Rightarrow \boxed{L = \frac{P}{2}} \quad (6)$$

Ex 012-020

Q3

Find Fourier series of the given

$f(x)$, which is assumed to have

period 2π . Show that you work

in detail.

Q2

$$f(x) = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x > 0 \end{cases}$$

(1)

$$f(x+2\pi) = f(x) \quad (2)$$

$$f(x+p) = f(x) \quad (3)$$

$$p=2\pi \quad (4)$$

$$\Rightarrow p=2\pi \quad (5)$$

$$L=\pi \quad (6)$$

Sol

Q1

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx) \right)$$

(7)

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \quad (8)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \quad (9)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \quad (10)$$

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$$a_0 = \frac{1}{2\pi} \left[\int_{-\pi}^0 (-x) dx + \int_0^{\pi} (x) dx \right] \quad (11)$$

$$a_0 = \frac{1}{2\pi} \left[-\frac{x^2}{2} \Big|_{-\pi}^0 + \frac{x^2}{2} \Big|_0^{\pi} \right]$$

$$a_0 = \frac{\pi}{2} \quad (12)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 (-x) \cos nx dx + \int_0^{\pi} x \cos nx dx \right]$$

$$a_n = \frac{1}{\pi} \left[- \left\{ x \frac{\sin nx}{n} \right\}_{-\pi}^0 - \int_{-\pi}^0 \frac{\sin nx}{n} dx \right]$$

$$+ \frac{1}{\pi} \left[x \frac{\sin nx}{n} \Big|_{-\pi}^0 - \int_{-\pi}^0 \frac{\sin nx}{n} dx \right]$$

(05)

$$\Rightarrow a_n = \frac{1}{\pi} \left[- \left\{ \frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right\}_{-\pi}^0 + \left\{ \frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right\}_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[- \left\{ \frac{n x \sin(nx) + \cos(nx)}{n^2} \right\}_{-\pi}^0 + \left\{ \frac{n x \sin nx + \cos nx}{n^2} \right\}_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[- \left\{ \frac{0+1}{n^2} - \frac{0+\cos(n\pi)}{n^2} \right\} + \left\{ \frac{0+\cos(n\pi)}{n^2} - \frac{0+1}{n^2} \right\} \right]$$

$$= \frac{1}{\pi} \left[-\frac{1}{n^2} (1 - \cos n\pi) + \frac{1}{n^2} (\cos n\pi - 1) \right]$$

$$= \frac{1}{\pi n^2} \left[\cos n\pi - 1 + \cos n\pi - 1 \right]$$

$$= \frac{1}{\pi n^2} \left[2 \cos n\pi - 2 \right]$$

$$\Rightarrow \boxed{a_n = \frac{2(\cos n\pi - 1)}{\pi n^2}}$$

Similarly $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 -x \sin nx \, dx + \int_0^{\pi} x \sin nx \, dx \right]$$

$$= 0 \quad (\text{Try to solve})$$

Thus

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad ; \quad \text{--- (1)}$$

$$\Rightarrow f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left(\frac{2(\cos n\pi - 1)}{n^2 \pi} \cdot \cos(nx) + 0 \right)$$

$$\Rightarrow \boxed{f(x) = |x| = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left(\frac{2(\cos n\pi - 1)}{n^2 \pi} \cdot \cos(nx) \right)} \quad \underline{\underline{\text{Ans}}}$$

Q.13: $f(x) = \begin{cases} x & \text{if } -\pi < x < 0 \\ \pi - x & \text{if } 0 < x < \pi \end{cases}$

— (1) $f(x+2\pi) = f(x)$

Sol: $f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$ — (2)

$P = 2\pi$
 $L = \pi$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \left[\int_{-\pi}^0 x dx + \int_0^{\pi} (\pi - x) dx \right] \quad \text{--- (3)}$$

$$= \frac{1}{2\pi} \left[\frac{x^2}{2} \Big|_{-\pi}^0 + \left(\pi x - \frac{x^2}{2} \right) \Big|_0^{\pi} \right]$$

$$= \frac{1}{2\pi} \left[0 - \frac{(-\pi)^2}{2} + \left\{ \left(\pi \cdot \pi - \frac{\pi^2}{2} \right) - 0 \right\} \right]$$

$$= \frac{1}{2\pi} \left[-\frac{\pi^2}{2} + \pi^2 - \frac{\pi^2}{2} \right] = \frac{1}{2\pi} \left[\frac{-\pi^2 + 2\pi^2 - \pi^2}{2} \right] = 0$$

$$\Rightarrow \boxed{a_0 = 0} \quad \text{--- (4)}$$

Remark:

$$\sin n\pi = 0$$

$$\cos n\pi = (-1)^n$$

$$a_n = \frac{1}{\lambda} \left(\int_{-\lambda}^{\lambda} f(x) dx \right) \quad \text{--- (5)}$$

$$= \frac{1}{\lambda} \left[\int_{-\lambda}^0 x \cos(nx) dx + \int_0^{\lambda} (\lambda - x) \cos nx dx \right] \quad \text{--- (6)}$$

$$= \frac{1}{\lambda} \left[\left. \frac{nx \sin nx + \cos nx}{n^2} \right|_{-\lambda}^0 + \left\{ (\lambda - x) \frac{\sin nx}{n} - \int \frac{(-1) \sin(nx)}{n} dx \right\} \right]_{\lambda}^{\lambda}$$

$$= \frac{1}{\lambda} \left[\frac{1}{n^2} (-\cos n\lambda + 1) + \left\{ \frac{(\lambda - x) \sin nx}{n} - \frac{\cos nx}{n^2} \right\} \right]_{\lambda}^{\lambda}$$

$$= \frac{1}{\lambda} \left[\frac{1}{n^2} (-\cos n\lambda + 1) + \left\{ (0 - \frac{\cos(n\lambda)}{n^2}) - (0 - \frac{1}{n^2}) \right\} \right]$$

$$= \frac{1}{\lambda} \left[\frac{1}{n^2} (-\cos n\lambda + 1) + \left\{ -\frac{\cos n\lambda}{n^2} + \frac{1}{n^2} \right\} \right]$$

$$= \frac{1}{\lambda} \left[\frac{1}{n^2} (-\cos n\lambda + 1 - \cos n\lambda + 1) \right]$$

$$\boxed{a_n = 0}$$

$$a_n = \frac{2}{n^2 \lambda} (1 - \cos(n\lambda))$$

$$b_n = \frac{1}{\lambda} \int_{-\lambda}^{\lambda} f(x) \sin(nx) dx = \frac{1}{\lambda} \left[\int_{-\lambda}^0 x \sin(nx) dx + \int_0^{\lambda} (\lambda - x) \sin nx dx \right]$$

$$= \frac{1}{\lambda} \left[\left. \frac{\sin(nx) - nx \cos nx}{n^2} \right|_{-\lambda}^0 + \left. \frac{n(x - \lambda) \cos nx - \sin nx}{n^2} \right|_0^{\lambda} \right]$$

$$= \frac{1}{\lambda} \left[\left(0 - \frac{\sin(-n\lambda) - n(-\lambda) \cos(-n\lambda)}{n^2} \right) + \left(0 - \frac{n(\lambda) \cos(0) - \sin(0)}{n^2} \right) \right]$$

$$= \frac{1}{\lambda} \left[-\frac{n\lambda \cos(n\lambda)}{n^2} + \frac{n\lambda}{n^2} \right] = \frac{\pi}{n\lambda} [1 - \cos(n\lambda)]$$

Q10

(5)

$$\Rightarrow b_n = \frac{1}{n} (1 - \cos n\pi) \quad \text{--- (7)}$$

$$\Rightarrow f(x) = 0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (8)}$$

$$\Rightarrow f(x) = \sum_{n=1}^{\infty} \left[\left(\frac{2}{n^2 \pi} (1 - \cos n\pi) \right) \cos nx + \left(\frac{1}{n} (1 - \cos n\pi) \right) \sin nx \right]$$

Q11

Exercise

Q1 $f(x) = \begin{cases} -x & \text{if } -\pi < x < 0 \\ x & \text{if } 0 < x < \pi \end{cases}, \quad f(x+2\pi) = f(x)$
 $P = 2\pi$

Q2 $f(x) = \begin{cases} x & \text{if } 0 < x < \pi \\ 0 & \text{if } -\pi < x < 0 \end{cases}, \quad f(x+\pi) = f(x)$

Q3 $f(x) = x + \pi, \quad f(x+2\pi) = f(x)$
 where $-\pi \leq x \leq \pi$

Q4 $f(x) = x^3, \quad f(x+2\pi) = f(x)$
 $-\pi \leq x \leq \pi$

Ex

Hints (09)

Find Fourier series of

$$f(x) = x^2 \quad -\pi < x < \pi; \quad f(x+2\pi) = f(x)$$

(1) (L = \pi)

Sol Hints

01

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

(2)

02

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{\pi^3}{3}$$

(3)

03

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx = \frac{4}{n^2} \cos n\pi$$

(4)

04

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx dx = 0$$

(5)

put (3), (4), (5) in (2).

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

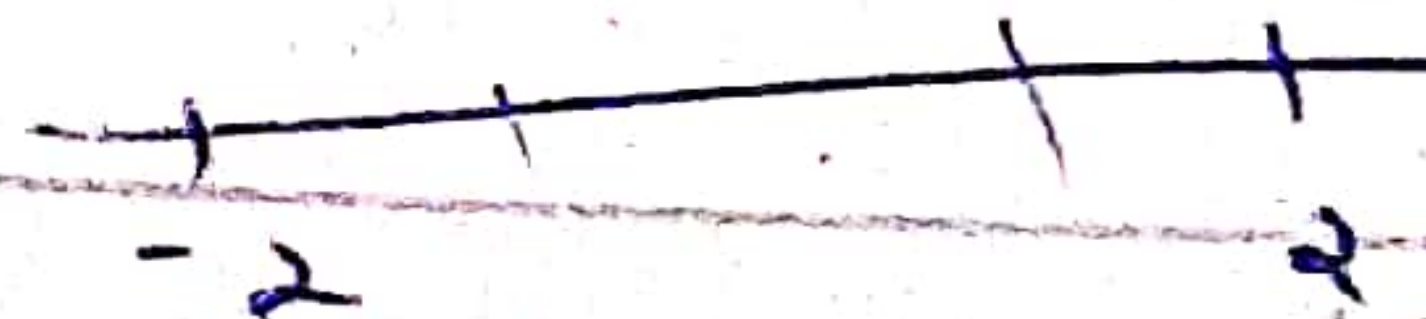
$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Exple Find the Fourier series of the function

$$f(x) = \begin{cases} 0, & \text{if } -2 < x < -1 \\ K, & \text{if } -1 < x < 1 \\ 0, & \text{if } 1 < x < 2 \end{cases}$$

$$p = 2L = 4, \quad L = 2$$



Sol: Since $f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right) \right)$

$$a_0 = \frac{1}{2(2)} \int_{-2}^2 f(x) dx = \frac{1}{4} \int_{-2}^{-1} 0 dx + \frac{1}{4} \int_{-1}^1 K dx + \frac{1}{4} \int_1^2 0 dx$$

$$a_0 = 0 + \frac{1}{4} kx \Big|_{-1}^1 + 0 = \frac{k}{4} (1 - (-1)) = \frac{2k}{4} = \frac{k}{2}$$

$$\Rightarrow \boxed{a_0 = \frac{k}{2}}$$

$$a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos\left(\frac{n\pi}{2}x\right) dx$$

$$= \frac{1}{2} \int_{-2}^{-1} 0 \cdot \cos\left(\frac{n\pi}{2}x\right) dx + \frac{1}{2} \int_{-1}^1 k \cos\left(\frac{n\pi}{2}x\right) dx + \frac{1}{2} \int_1^2 0 \cdot \cos\left(\frac{n\pi}{2}x\right) dx$$

$$= 0 + \frac{k}{2} \left(\frac{\sin\left(\frac{n\pi}{2}x\right)}{\frac{n\pi}{2}} \right) \Big|_{-1}^1 + 0$$

$$= \frac{2k}{2n\pi} \left[\sin\left(\frac{n\pi}{2}\right) - \sin\left(-\frac{n\pi}{2}\right) \right]$$

$$= \frac{k}{n\pi} \left[\sin\left(\frac{n\pi}{2}\right) + \sin\left(\frac{n\pi}{2}\right) \right]$$

0, 1, 0, 1, 0, 1, ...

$$\boxed{a_n = \frac{2k}{n\pi} \sin\left(\frac{n\pi}{2}\right)}$$

Thus $a_n = 0$ if n is even

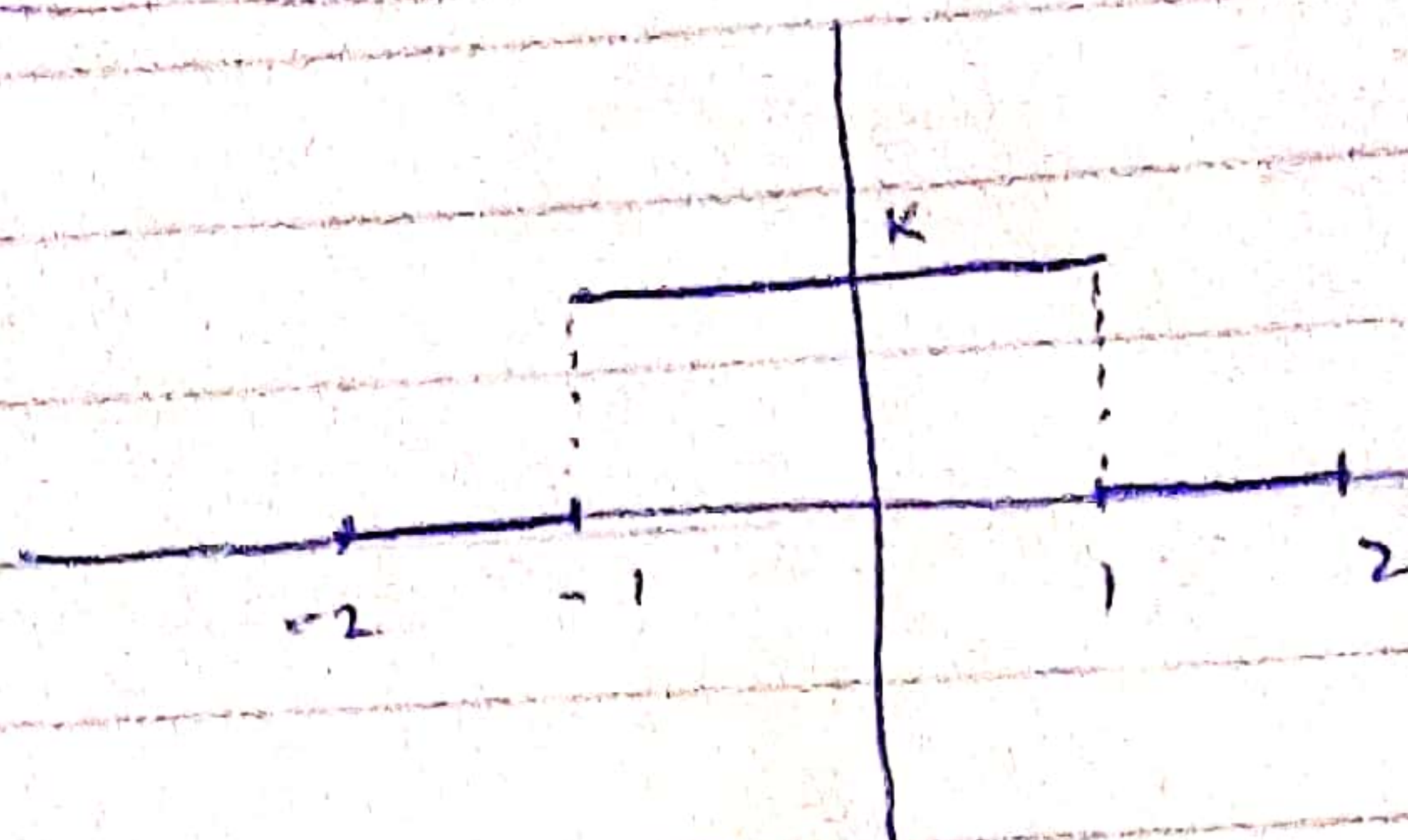
$$a_n = \frac{2k}{n\pi} \text{ if } n = 1, 5, 9, \dots$$

$$a_n = -\frac{2k}{n\pi} \text{ if } n = 3, 7, 11, \dots$$

Also $b_n = \frac{1}{2} \int_{-2}^2 f(x) \sin\left(\frac{n\pi}{2}x\right) dx = 0 \Rightarrow \boxed{b_n = 0}$

$$\text{Thus } f(x) = \frac{k}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{2}x\right)$$

$$\Rightarrow f(x) = \frac{k}{2} + \frac{2k}{\pi} \left(\cos \frac{\pi}{2} x - \frac{1}{3} \cos \frac{3\pi}{2} x + \frac{1}{5} \cos \frac{5\pi}{2} x - \dots \right)$$



Exp-2: (Periodic Rectangular Wave, change of scale)
Find the "Fourier series" of the function

$$f(x) = \begin{cases} -k & \text{if } -2 < x < 0 \\ k & \text{if } 0 < x < 2 \end{cases}, p=2L=4, L=2.$$

(Solve it)

Simplification: Even and odd functions: If $f(x)$ is an

even function, that is $f(-x) = f(x)$, its Fourier series reduces to a Fourier cosine series

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L} x\right) \quad (\because f \text{ even})$$

with $a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$, $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L} x\right) dx$; $n=1, 2, \dots$

Similarly if $f(x)$ is even odd function then ($f(-x) = -f(x)$)
Then

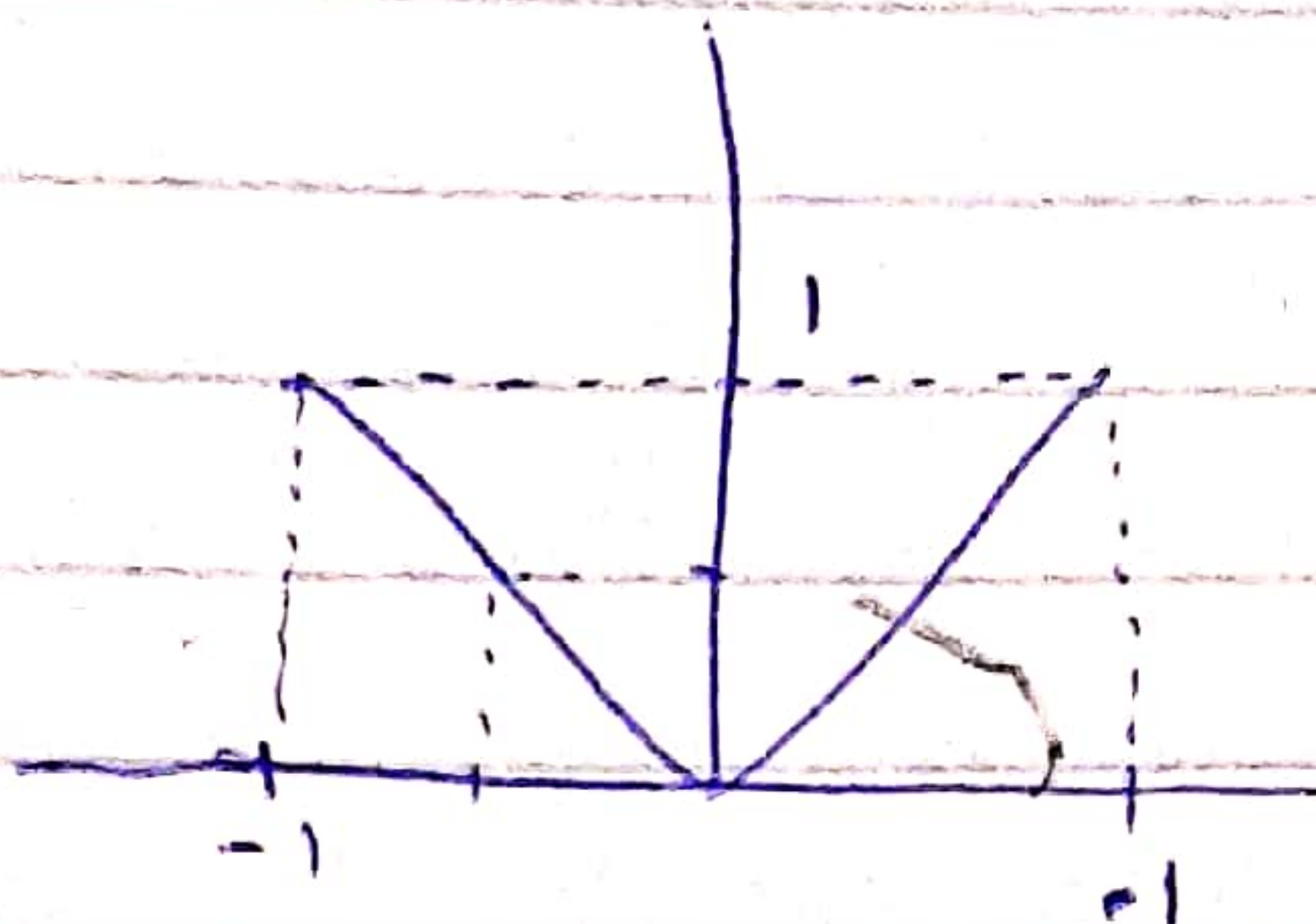
$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L} x\right) \quad (\because f \text{ odd})$$

with $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$

Exercise: 11.2 Fourier Series for period $P=2L$

Is the given function even or odd or neither even nor odd? Find its Fourier series. Show details of your work (Q.8 - Q.17)

Q.8:



Sol:

$$f(x) = \begin{cases} -x & ; -1 < x < 0 \\ x & ; 0 < x < 1 \end{cases}$$

The function is an even function bcz

$$f(-1) = -(-1) = 1 ; f(-1/2) = -(-1/2) = 1/2$$

$$\text{Also } f(1) = 1 , f(1/2) = 1/2$$

Fourier series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right)$$

$$\text{or } a_0 = \frac{1}{L} \int_0^L f(x) dx$$

$$\begin{aligned} \Rightarrow a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx = \frac{1}{2} \int_{-1}^1 f(x) dx = \frac{1}{2} \int_{-1}^0 -x dx + \frac{1}{2} \int_0^1 x dx \\ &= -\frac{1}{2} \left. \frac{x^2}{2} \right|_{-1}^0 + \frac{1}{2} \left. \frac{x^2}{2} \right|_0^1 = -\frac{1}{4} (0+1) + \frac{1}{4} (1-0) = 0 \end{aligned}$$

$$\Rightarrow \boxed{a_0 = 0}$$

$$a_n = \frac{1}{2} \int_{-1}^1 f(x) \cos(n\pi x) dx ; n=1,2, \dots$$

$$\Rightarrow a_n = \int_{-1}^0 -x \cos(n\pi x) dx + \int_0^1 x \cos(n\pi x) dx$$

$$\Rightarrow a_n = \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$$

Try to solve and then put a_0 and a_n

in $\boxed{f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\pi x) ; n=1,2, \dots}$

Ans

X